PROPOSED SOLUTION ARCHITECTURE

Developer: [Left Blank] | OptiCrop Project Documentation
Repository: https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine

1. System Concept

OptiCrop is an agricultural optimization system that combines machine learning pipelines with an interactive web portal. It runs three models in parallel (Random Forest Classifier, Random Forest Regressor, and K-Means Clustering) to process soil parameters and deliver recommendations to users in real time.

Opticrop - Proposed Solution

An AI-driven Agricultural Optimizer delivering five core features:

1. Crop Recommender - RF classifier, 22 crop classes, 100% test accuracy
2. Yield Forecaster - RF regressor predicts final yield in t/ha
3. Soil Diagnoser - K-Means clustering into 5 soil health clusters
4. NPK Prescriber - deficit-based Urea / SSP / MOP amendment guide
5. Analytics Dashboard- accuracy charts, confusion matrix, feature importance

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2. High-Level System Features

- Agricultural Form: Form to input soil parameters, region, and season.
- Analysis Panel: Displays the recommended crop, confidence, yield forecast, and soil health classification.
- NPK Guidance Card: Provides chemical corrective recommendations (e.g., urea, SSP, MOP) based on soil values.
- Telemetry Dashboard: Displays classifier performance comparisons, Gini feature rankings, and a confusion matrix heatmap.